



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO DD FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

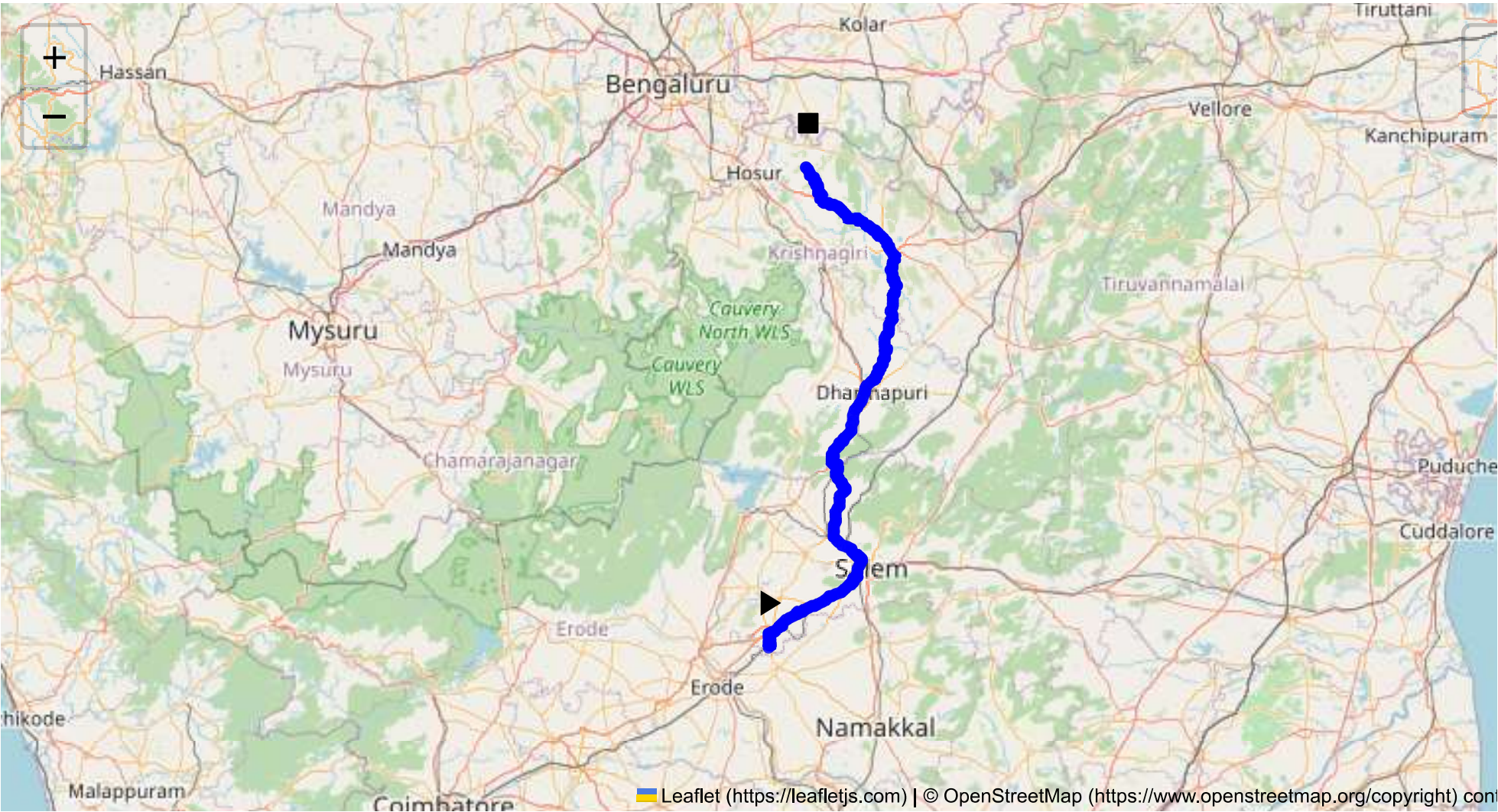
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 191.39 km**  
**Estimated Duration: 3.5 hours**  
**Adjusted Duration (Heavy Vehicle): 4.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.750149, 77.975897)**



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The journey from Sangagiri, Tamil Nadu to Athimugam, Tamil Nadu covers approximately 191.39 kilometers, primarily along National Highways and state roads. The route typically involves the NH44 and several smaller state roads that connect major truck routes.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu experiences a tropical climate with high temperatures year-round. The monsoon season occurs from October to December, which can bring heavy rains.
- **Hazards:** During the monsoon, road conditions can deteriorate due to flooding, landslides in hilly areas, and reduced visibility, increasing the risk of accidents.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Peak traffic is usually observed in the early morning (7-9 AM) and late evening (5-8 PM) due to commuter and local business traffic.
- **Congestion-Prone Areas:** Hosur's entry points are known for congestion, particularly at industrial areas and markets along NH44.

## 4. Assessment of Road Quality and Infrastructure

- Major highways like NH44 are generally well-maintained, but smaller state roads may have potholes and uneven surfaces, especially after monsoon seasons.
- Frequent road maintenance and construction work could lead to temporary detours and disruptions.

## 5. Suggestions for Alternative Routes

- In case of emergencies or road blockages, using the NH79 to reach NH44 offers an alternative, albeit possibly longer, path as it connects various critical junctions.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials requires special permits in Tamil Nadu.
- Regulations may restrict passage through ecologically sensitive zones and urban areas during peak hours.

## 7. Overview of Historical Incidents

- There have been reported incidents related to chemical spills and vehicular fires on NH44, especially when passing through densely populated regions near urban centers.

## 8. Environmental Considerations and Sensitive Areas

- Several zones along the route, such as river crossings and forest reserves, necessitate careful driving and adherence to environmental protection protocols.

## 9. Analysis of Communication Coverage

- Major highways like NH44 generally have good mobile network coverage.
- Potential communication dead zones are likely along some interior state roads and rural stretches.

## 10. Estimated Emergency Response Times

- Near urban centers like Hosur, emergency services can reach within 15-30 minutes.
- In rural or remote areas, response times could extend to 1 hour or more due to distance and road accessibility.

## 11. Overall Summary of Risk Assessment

- **Strengths:** Major highways facilitate quicker travel and better road conditions.
- **Weaknesses:** Smaller state roads can pose hazards due to poor maintenance and increased traffic during monsoon or festivals.
- **Risks:** Weather-related events and historical incidents necessitate heightened caution for hazardous material transport.
- **Recommendations:** Constant weather monitoring, adherence to transport regulations, and alternative route planning are essential for safe travel. Additionally, maintaining communication equipment to navigate potential dead zones is critical.

Overall, while the route provides efficient access between key locations, drivers must remain vigilant regarding weather impacts, road conditions, and traffic regulations to mitigate risks effectively.

### Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 0  | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn      | Medium     | 11.43995, 77.87431 | 30 KM/Hr    |
| 2  | Turn      | High       | 11.44107, 77.87528 | 15 KM/Hr    |
| 3  | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 4  | Turn      | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 5  | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 6  | Turn      | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 7  | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 8  | Turn      | Medium     | 11.47986, 77.88397 | 30 KM/Hr    |
| 9  | Turn      | High       | 11.53606, 77.95869 | 15 KM/Hr    |
| 10 | Turn      | Medium     | 11.62403, 78.10893 | 30 KM/Hr    |
| 11 | Turn      | Medium     | 11.62932, 78.11659 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 12 | Turn      | High       | 11.64183, 78.11810 | 15 KM/Hr    |
| 13 | Turn      | Medium     | 11.64338, 78.11998 | 30 KM/Hr    |
| 14 | Turn      | Medium     | 11.67903, 78.12796 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.69532, 78.10764 | 30 KM/Hr    |
| 16 | Turn      | Medium     | 11.73968, 78.05098 | 30 KM/Hr    |
| 17 | Turn      | Medium     | 11.86790, 78.08347 | 30 KM/Hr    |
| 18 | Turn      | Medium     | 11.91357, 78.05868 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.92556, 78.06499 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.93296, 78.06432 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.93535, 78.06150 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.93694, 78.05425 | 30 KM/Hr    |
| 23 | Turn      | Medium     | 11.95531, 78.04698 | 30 KM/Hr    |
| 24 | Turn      | High       | 11.95803, 78.04964 | 15 KM/Hr    |
| 25 | Turn      | Medium     | 11.96068, 78.04960 | 30 KM/Hr    |
| 26 | Turn      | High       | 11.96370, 78.05142 | 15 KM/Hr    |
| 27 | Turn      | High       | 11.96650, 78.04927 | 15 KM/Hr    |
| 28 | Turn      | Medium     | 12.22422, 78.19127 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 12.24685, 78.19561 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 12.25012, 78.19946 | 30 KM/Hr    |
| 31 | Turn      | Medium     | 12.25280, 78.20041 | 30 KM/Hr    |
| 32 | Turn      | Medium     | 12.29065, 78.19892 | 30 KM/Hr    |
| 33 | Turn      | Medium     | 12.32012, 78.20734 | 30 KM/Hr    |
| 34 | Turn      | Medium     | 12.40099, 78.21805 | 30 KM/Hr    |
| 35 | Turn      | Medium     | 12.45136, 78.21799 | 30 KM/Hr    |
| 36 | Turn      | Medium     | 12.50485, 78.22336 | 30 KM/Hr    |
| 37 | Turn      | Medium     | 12.59710, 78.13638 | 30 KM/Hr    |
| 38 | Turn      | Medium     | 12.61895, 78.08313 | 30 KM/Hr    |
| 39 | Turn      | Medium     | 12.63692, 78.07068 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 12.66132, 78.01539 | 30 KM/Hr    |
| 41 | Turn      | High       | 12.67002, 78.01671 | 15 KM/Hr    |
| 42 | Turn      | Medium     | 12.67454, 78.01437 | 30 KM/Hr    |
| 43 | Turn      | High       | 12.67572, 78.01196 | 15 KM/Hr    |
| 44 | Turn      | Medium     | 12.70398, 78.00088 | 30 KM/Hr    |
| 45 | Turn      | High       | 12.71385, 77.99942 | 15 KM/Hr    |

|    |           |            |                    |             |
|----|-----------|------------|--------------------|-------------|
|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
| 46 | Turn      | Medium     | 12.71574, 77.99671 | 30 KM/Hr    |

## Emergency Locations

|    |          |                                                           |                           |             |            |
|----|----------|-----------------------------------------------------------|---------------------------|-------------|------------|
|    | type     | name                                                      | coordinates               | speed_limit | risk_level |
| 0  | hospital | Thilagam Hospital                                         | 11.4753573,<br>77.8685829 | 30 km/h     | Medium     |
| 1  | clinic   | Meyyazhagan Clinic                                        | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2  | hospital | Dr. Jaganathan Hospital                                   | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital                                       | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                                           | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | clinic   | Dr. Dhanasekar Clinic                                     | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 6  | hospital | Sri Vellaiyan Hospital                                    | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 8  | hospital | C. N. Hospital                                            | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 9  | hospital | Annapoorna Medical College Hospital                       | 11.57693,<br>78.037627    | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 12 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College    | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 17 | hospital | Bharathi Hospital, Salem                                  | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 18 | hospital | Sisu Hospital Inida Ltd                                   | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 19 | hospital | Aishwarya Hospital                                        | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 21 | hospital | C.M. Hospital                                             | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 22 | pharmacy | Healthy Pharmacy                                          | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |

|    | type     | name                                      | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------------------------|---------------------------|-------------|------------|
| 23 | pharmacy | Thulasi Pharmacy                          | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 24 | hospital | Sri Gokulam Speciality Hospital           | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 25 | hospital | Global Medical Center                     | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 26 | hospital | Global Medical Centre<br>Jagirammapalayam | 11.6894409,<br>78.1108847 | 30 km/h     | Medium     |
| 27 | hospital | Manipal Hospital                          | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 31 | clinic   | Manimuthu Skin Clinic                     | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 32 | hospital | St. Joseph's Speciality Hospital          | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 35 | hospital | L.R. Hospital                             | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 36 | clinic   | JSA Clinic                                | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 38 | hospital | Government Hospital                       | 12.3048002,<br>78.2060881 | 30 km/h     | Medium     |
| 43 | hospital | TCR Multi Speciality Hospital             | 12.5088183,<br>78.2237657 | 30 km/h     | Medium     |
| 44 | hospital | St. Mary's Hospital                       | 12.5128555,<br>78.2186887 | 30 km/h     | Medium     |
| 45 | hospital | KVS Hospital                              | 12.5277857,<br>78.2055049 | 30 km/h     | Medium     |
| 46 | hospital | Ga Heart Hospital                         | 12.5287708,<br>78.2055336 | 30 km/h     | Medium     |
| 47 | clinic   | Psychiatric Centre                        | 12.529476,<br>78.205064   | 30 km/h     | Medium     |
| 48 | clinic   | Shree Ramachandra ENT Centre              | 12.5290096,<br>78.2037359 | 30 km/h     | Medium     |
| 54 | hospital | Shoolagiri Government Hospital            | 12.6669051,<br>78.015584  | 30 km/h     | Medium     |



## Crowded Spots

|    | type       | name                                                    | coordinates            | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|------------------------|-------------|------------|
| 11 | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662, 78.0839876 | 30 km/h     | Medium     |
| 13 | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056, 78.0843617 | 30 km/h     | Medium     |
| 14 | school     | Dist Education Training                                 | 11.6133799, 78.1022409 | 30 km/h     | Medium     |
| 15 | school     | DIET - Salem                                            | 11.6144158, 78.1025078 | 30 km/h     | Medium     |
| 16 | school     | Diamond Rays Matric HR Sec School                       | 11.6385629, 78.1206259 | 30 km/h     | Medium     |
| 20 | college    | Sona College                                            | 11.6768, 78.1257052    | 30 km/h     | Medium     |
| 28 | college    | Govt Engineering College                                | 11.7128537, 78.0877084 | 30 km/h     | Medium     |
| 29 | university | Periyar University                                      | 11.7186758, 78.0771089 | 30 km/h     | Medium     |
| 30 | school     | Fathima Hr Sec School                                   | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 33 | school     | John Britto Matriculation Higher Secondary School       | 11.7676907, 78.0513092 | 30 km/h     | Medium     |
| 34 | college    | Dhirajlal Gandhi College of Technology                  | 11.7968014, 78.0562095 | 30 km/h     | Medium     |
| 37 | college    | Laxminarayana Polytechnic College                       | 12.0421277, 78.103203  | 30 km/h     | Medium     |
| 39 | school     | Government Boys Higher secondary school                 | 12.3047059, 78.2065495 | 30 km/h     | Medium     |
| 40 | school     | Nalanda Metric Higher Secondary School                  | 12.488407, 78.21856    | 30 km/h     | Medium     |
| 41 | school     | Nalanda International Public School                     | 12.4994979, 78.2235094 | 30 km/h     | Medium     |
| 42 | school     | Don Bosco School Ground                                 | 12.505954, 78.2222003  | 30 km/h     | Medium     |
| 49 | school     | Sri Saraswathi Vidyalaya                                | 12.5620799, 78.183049  | 30 km/h     | Medium     |
| 50 | college    | Government Medical College                              | 12.582332, 78.1496363  | 30 km/h     | Medium     |
| 51 | school     | Crescent Matriculation Higher Seconddary School         | 12.6003724, 78.132787  | 30 km/h     | Medium     |